



Descriptive Stats Primer

Descriptive Statistics: Measures of Central Tendency

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Ethical research involves mitigating risks to both **individuals and collectives**, in both the **short-term and long-term**

Qualitative and Quantitative research involve similar ethical challenges.

- Ethical concerns emerge at all stages of the research process: Data Collection, Storage, Analysis, Sharing, Interpretation, Policy Implications, etc.

Even 'desk researchers' need to prioritize ethics:

- Was the data collected ethically in the first place?
- Are we being transparent about the data sources, risks, and benefits?
- Do we have a responsibility to include an 'ethics statement'?
- Can the conclusions we draw generate harm?

What are descriptive statistics and why do we need them?

Measures of Central Tendency

Mean: the average value of the variable (the sum of all of the values divided by the total number values).

Median: the middle value of the variable.

- If the total number of obs is *odd*, it's the middle value.
- If the total number of obs is *even*, it's the mean of the middle two values.

Mode: the most common value of the variable.

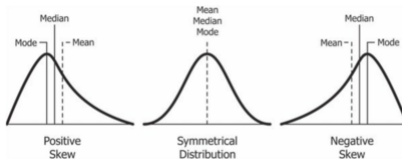
Mean and median are much more useful for continuous/interval data (sometimes even ordinal data with lots of categories).

Mode can be useful for categorical variables (because the categories are whole numbers and other measures may not produce whole numbers!)

Measures of Dispersion

Skewness:

- The median is preferable to the mean for highly skewed data.
- Positive (right) skew:
 - $\text{mean} > \text{median}$
- Negative (left) skew:
 - $\text{mean} < \text{median}$



Kurtosis:

- Positive (leptokurtic) skew: tall.
- Negative (platykurtic) skew: short.

